



DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECT PROPOSAL

1. Project Title: - Human Activity Recognition (HAR) – Battery-Efficient Personalized Model

2. Project Scope: -

This project aims to develop and validate a battery-efficient, personalized Human Activity Recognition (HAR) model for resource-constrained mobile and wearable devices. By leveraging feature engineering and subject-specific modeling techniques, the solution prioritizes accuracy alongside low power consumption, enabling practical deployment in real-world health monitoring, smart fitness, and IoT-based activity tracking scenarios.

In future implementations, several promising directions can further advance the scope and impact of this research:

- **On-device Real-time Deployment:** Transitioning the model from laboratory benchmarking to real-time operation on commercial wearables and smartphones via frameworks like TinyML or TensorFlow Lite will enable ubiquitous human activity monitoring with minimal latency and energy overhead. This will facilitate seamless integration into health, rehabilitation, and lifestyle apps tailored for individual users.
- **Transfer Learning and Automatic Personalization:** Incorporating advanced transfer learning strategies will allow models to rapidly adapt to new users and environments with minimal calibration data. Auto-personalization pipelines can be developed so that models continually refine themselves based on incoming user-specific data, further boosting prediction accuracy and long-term usability.
- **Privacy-Preserving Data Handling:** As HAR systems become increasingly embedded in everyday life, user data privacy and consent will grow in importance. Exploring edge-computing and federated learning approaches will limit the need for raw data transmission and support compliance with data protection regulations.
- **Robustness Across Diverse Activities:** Enlarging the dataset to feature more varied activities, sensor modalities (such as heart rate, GPS, and ambient sensors), and demographic groups will help improve the generalizability of the model in heterogeneous populations and use-cases.
- **Adaptive Battery Management:** Integrating smart battery prediction or adaptive sampling mechanisms will allow HAR algorithms to dynamically vary computation intensity based on device power state, user activity context, or desired monitoring frequency.
- **Commercialization and Societal Impact:** Future research can investigate deployment in large-scale commercial fitness bands, elderly care monitoring, workplace safety, and smart

city infrastructure, unlocking benefits such as personalized rehabilitation, preventive healthcare, and early anomaly detection.

In summary, this work lays the foundation for scalable, personalized, and sustainable activity recognition on low-power devices. Expanding on-device automation, privacy safeguards, and adaptive modeling will further enhance real-world impact and accelerate adoption across sectors.

3. Requirements: -

➤ Hardware Requirements

1. Wearable or Mobile Sensor Device:
2. Computing Device for Model Training & Evaluation

➤ Software Requirements

1. Jupyter Notebook
2. Git/GitHub
3. Android Studio

STUDENTS DETAILS

Name	UID	Signature
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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)